## SQL Query Implementation for Purple Company E-Commerce Database

## Project Title: Purple Company – Beauty Products E-Commerce Order Management System

# 1. Basic SQL Queries

## 1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.

### Display all products

```
SQL> SELECT * FROM PurpleProduct;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRICE      STOCK
-------------------------------------------------- ---------- ----------
CATEGORY_ID
-----------
         1
Matte Lipstick
Purple                                               499         50
          1


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRICE      STOCK
-------------------------------------------------- ---------- ----------
CATEGORY_ID
-----------
         2
Liquid Foundation
Purple                                               799         40
          2


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRICE      STOCK
-------------------------------------------------- ---------- ----------
CATEGORY_ID
-----------
         3
Waterproof Eyeliner
```

## Display selected product details

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         1
Matte Lipstick
       499

         2
Liquid Foundation
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         3
Waterproof Eyeliner
       349

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

         5
Volume Mascara
       449
```

## 1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price > 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         1
Matte Lipstick
       499

         2
Liquid Foundation
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         3
Waterproof Eyeliner
       349

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
       599

         5
Volume Mascara
       449
```

**Find products with price greater than 100**

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price > 100;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
         1
Matte Lipstick
        499

         2
Liquid Foundation
        799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------

         3
Waterproof Eyeliner
        349

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
     PRICE
----------
        599

         5
Volume Mascara
        449
```

**1.3 ORDER BY**

The ORDER BY clause is used to sort the query results.

## Products from lowest to highest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         3
Waterproof Eyeliner
       349

         5
Volume Mascara
       449

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         1
Matte Lipstick
       499

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599
```

## Products from highest to lowest price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         2
Liquid Foundation
       799

         4
Glow Blush
       599

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         1
Matte Lipstick
       499

         5
Volume Mascara

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       449

         3
Waterproof Eyeliner
       349
```

## 1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

**Display unique categories**

```
SQL> SELECT DISTINCT Category_ID
  2  FROM PurpleProduct;

CATEGORY_ID
-----------
          1
          2
          3
          4
          5
```

## 2. Search Products

## 2.1 Search Products Based on Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price > 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
          1
Matte Lipstick
       499

          2
Liquid Foundation
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

          3
Waterproof Eyeliner
       349

          4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

          5
Volume Mascara
       449
```

This displays products whose price is between ₹100 and ₹400.

## 2.2 Search Products Based on Category

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price > 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         1
Matte Lipstick
       499

         2
Liquid Foundation
       799

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         3
Waterproof Eyeliner
       349

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

         5
Volume Mascara
       449
```

This displays products belonging to category 101.

## 3. Retrieve Customer

Information Display all customers

```
SQL> SELECT * FROM PurpleCustomer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
         1
Arun
arun@gmail.com
9876543210


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
         2
Kavin
kavin@gmail.com
9876543211


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
         3
Priya
priya@gmail.com
9876543212
```

**Search for a particular customer**

```
SQL> SELECT Customer_ID, Customer_Name, Email, Phone
  2  FROM PurpleCustomer
  3  WHERE Customer_Name = 'Priya';

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
           3
Priya
priya@gmail.com
9876543212
```

## 4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

## 4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM PurpleProduct
  3  WHERE Price > 300 AND Stock > 10;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
         1
Matte Lipstick
       499         50

         2
Liquid Foundation
       799         40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

         3
Waterproof Eyeliner
       349         60

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       599         35

         5
Volume Mascara
       449         45
```

## 4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM PurpleProduct
  3  WHERE Price < 300 OR Stock > 20;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
         1
Matte Lipstick
       499         50

         2
Liquid Foundation
       799         40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

         3
Waterproof Eyeliner
       349         60

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       599         35
```

## 4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM PurpleProduct
  3  WHERE Price < 300 OR Stock > 20;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
         1
Matte Lipstick
       499         50

         2
Liquid Foundation
       799         40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------

         3
Waterproof Eyeliner
       349         60

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE      STOCK
---------- ----------
       599         35

         5
Volume Mascara
       449         45
```

## 4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price, Category_ID
  2  FROM PurpleProduct
  3  WHERE Category_ID IN (1, 2);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
         1
Matte Lipstick
       499           1

         2
Liquid Foundation
       799           2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------


SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  WHERE Price BETWEEN 300 AND 500;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         1
Matte Lipstick
       499

         3
Waterproof Eyeliner
```

## 5. Basic Business Reports

## 5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM PurpleProduct
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         3
Waterproof Eyeliner
       349

         5
Volume Mascara
       449

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         1
Matte Lipstick
       499

         4
Glow Blush

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       599

         2
Liquid Foundation
       799
```

## 5.2 Customer Order Report

```
SQL> SELECT * FROM PurpleCustomer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
          1
Arun
arun@gmail.com
9876543210


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
          2
Kavin
kavin@gmail.com
9876543211


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
          3
Priya
priya@gmail.com
9876543212
```


```
SQL> SELECT * FROM PurpleCustomer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
          1
Arun
arun@gmail.com
9876543210


CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE
---------------
          2
Kavin
kavin@gmail.com
9876543211
```


## 5.3 Product Order Report

```
SQL> SELECT Product_ID, Product_Name
  2  FROM PurpleProduct;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
         1
Matte Lipstick

         2
Liquid Foundation

         3
Waterproof Eyeliner


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
         4
Glow Blush

         5
Volume Mascara
```

## 5.4 Customer-wise Total Orders

```
SQL> SELECT C.Customer_ID, C.Customer_Name,
  2         COUNT(O.Order_ID) AS Total_Orders
  3  FROM PurpleCustomer C
  4  LEFT JOIN PurpleOrder O
  5  ON C.Customer_ID = O.Customer_ID
  6  GROUP BY C.Customer_ID, C.Customer_Name
  7  ORDER BY C.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
         1
Arun
          0

         2
Kavin
          0

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------

         3
Priya
          0

         4
Anu

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
          0
```

## 5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         NVL(SUM(OI.Quantity * OI.Unit_Price), 0)
  4         AS Total_Purchase_Amount
  5  FROM PurpleCustomer C
  6  LEFT JOIN PurpleOrder O
  7  ON C.Customer_ID = O.Customer_ID
  8  LEFT JOIN PurpleOrderItem OI
  9  ON O.Order_ID = OI.Order_ID
 10  GROUP BY C.Customer_ID, C.Customer_Name
 11  ORDER BY C.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
          1
Arun
                    0

          2
Kavin
                    0

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------

          3
Priya
                    0

          4
Anu
```